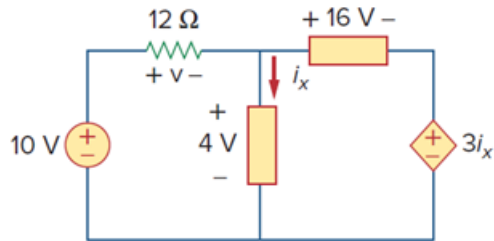


Problem

Calculate v and i_x in the circuit of Fig. 2.79.

Figure 2.79:



Step-by-step solution

Step 1 of 4

The circuit is shown in Figure 1.

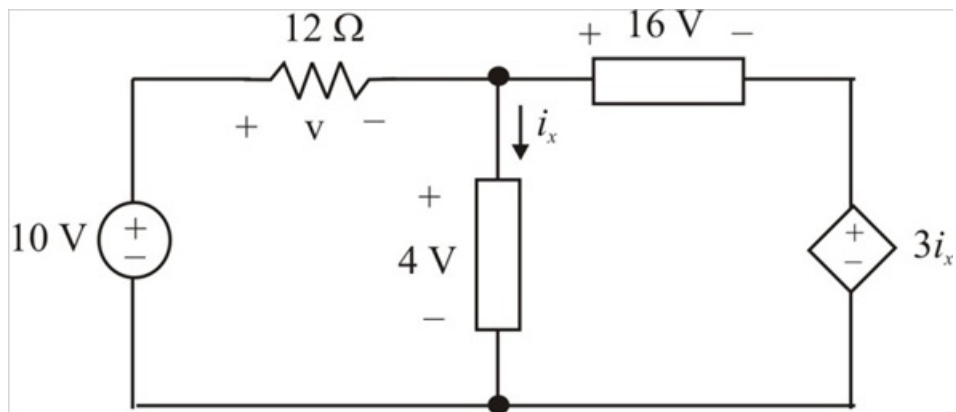


Figure 1

Step 2 of 4

Apply loops to the circuit as shown in Figure 2.

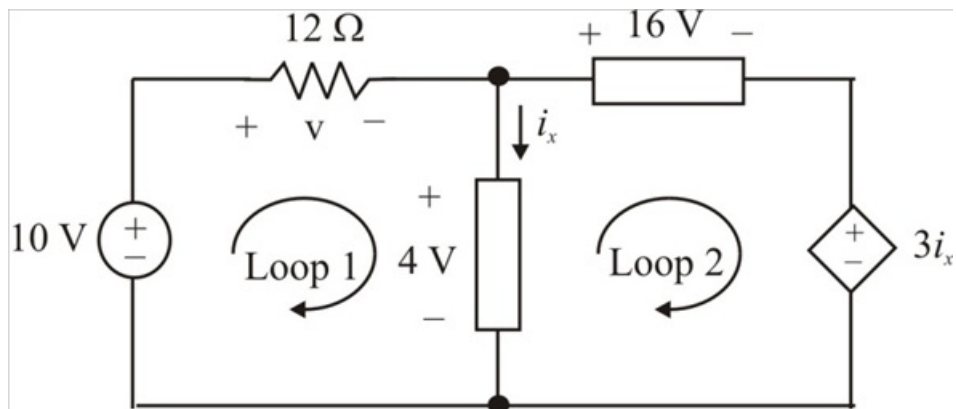


Figure 2

Step 3 of 4

Apply Kirchhoff's voltage law, the algebraic sum of all voltages around a closed path (or loop) is zero. Calculate the voltage v .

$$-10 + v + 4 = 0$$

$$v = 6 \text{ V}$$

Therefore, the voltage v is .

Step 4 of 4

Apply Kirchhoff's voltage law, calculate the current i_x .

$$-4 + 16 + 3i_x = 0$$

$$i_x = -4 \text{ A}$$

Therefore, the current i_x is .

[Comments \(1\)](#)



Anonymous

Is 3ix voltage???
